

Output Q1 - Top 3 Highest-Earning Employees per Department

MySQL 8.0 Command Line Client

```
mysql> SELECT d.dept_name,  
->         e.employee_name,  
->         e.salary  
-> FROM Employees e  
-> INNER JOIN Departments d  
-> ON e.dept_id = d.dept_id  
-> WHERE e.salary IN  
-> (  
->     SELECT DISTINCT e2.salary  
->     FROM Employees e2  
->     WHERE e2.dept_id = e.dept_id  
->     ORDER BY e2.salary DESC  
->     LIMIT 3  
-> )  
-> ORDER BY d.dept_name, e.salary DESC;
```

```
+-----+-----+-----+  
| dept_name | employee_name | salary   |  
+-----+-----+-----+  
| HR        | Neha Joshi    | 70000.00 |  
| HR        | Rohan Das     | 68000.00 |  
| HR        | Kavita Menon  | 65000.00 |  
| IT        | Vikram Rao    | 120000.00 |  
| IT        | Ananya Iyer   | 115000.00 |  
| IT        | Karan Mehta   | 110000.00 |  
| Sales     | Rahul Sharma  | 95000.00  |  
| Sales     | Priya Singh   | 88000.00  |  
| Sales     | Amit Patel    | 82000.00  |  
+-----+-----+-----+  
9 rows in set (0.00 sec)
```

Output Q2 - View: Department Salary Statistics

MySQL 8.0 Command Line Client

```
mysql> CREATE VIEW vw_dept_salary_stats AS
-> SELECT d.dept_name,
->        COUNT(e.employee_id) AS total_employee_count,
->        SUM(e.salary) AS total_salary_expenditure,
->        AVG(e.salary) AS average_salary
-> FROM Departments d
-> INNER JOIN Employees e
-> ON d.dept_id = e.dept_id
-> GROUP BY d.dept_name;
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> SELECT * FROM vw_dept_salary_stats;
```

dept_name	total_employee_count	total_salary_expenditure	average_salary
HR	3	203000.00	67666.67
IT	4	443000.00	110750.00
Sales	4	340000.00	85000.00

3 rows in set (0.00 sec)

Output Q3 - User-Defined Function: Net Salary

MySQL 8.0 Command Line Client

```
mysql> DELIMITER //
```

```
mysql> CREATE FUNCTION fn_net_salary(  
  ->    basic_pay DECIMAL(10,2),  
  ->    bonuses DECIMAL(10,2),  
  ->    deductions DECIMAL(10,2)  
  -> )  
  -> RETURNS DECIMAL(10,2)  
  -> DETERMINISTIC  
  -> RETURN basic_pay + bonuses - deductions //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> DELIMITER ;
```

```
mysql> SELECT fn_net_salary(50000, 5000, 2000) AS net_salary;
```

net_salary
53000.00

1 row in set (0.00 sec)

Output Q4 - Trigger: Log Salary Changes

MySQL 8.0 Command Line Client

```
mysql> DELIMITER //
```

```
mysql> CREATE TRIGGER trg_log_salary_change
```

```
-> AFTER UPDATE ON Employees
```

```
-> FOR EACH ROW
```

```
-> BEGIN
```

```
->     IF OLD.salary <> NEW.salary THEN
```

```
->         INSERT INTO EmpHistory
```

```
->             (employee_id, old_salary, new_salary, change_date)
```

```
->             VALUES
```

```
->             (OLD.employee_id, OLD.salary, NEW.salary, NOW());
```

```
->     END IF;
```

```
-> END //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> DELIMITER ;
```

```
mysql> UPDATE Employees SET salary = 125000.00 WHERE employee_id = 5;
```

Query OK, 1 row affected (0.01 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
mysql> SELECT employee_id, old_salary, new_salary, change_date
```

```
-> FROM EmpHistory WHERE employee_id = 5;
```

employee_id	old_salary	new_salary	change_date
5	120000.00	125000.00	2026-09-03 11:42:07

1 row in set (0.00 sec)